direction connecting the centroid to the query point, that in the general case won't lie on any of the axes.

A different approach to reduce the wasted area attempts to reduce the volume of the bounding spheres used: so far we have always used spheres whose center was a group of points' center of mass, and whose radius was the distance to the furthest point, so that the sphere would cover all points in the cluster. This, however, is not the smallest possible sphere that covers all the points: figure 10.26 shows an example of such difference.

Computing this smallest enclosing sphere in higher dimensions is, however, not feasible, because the algorithm to compute the exact values for its center (and radius) is exponential in the number of dimensions.

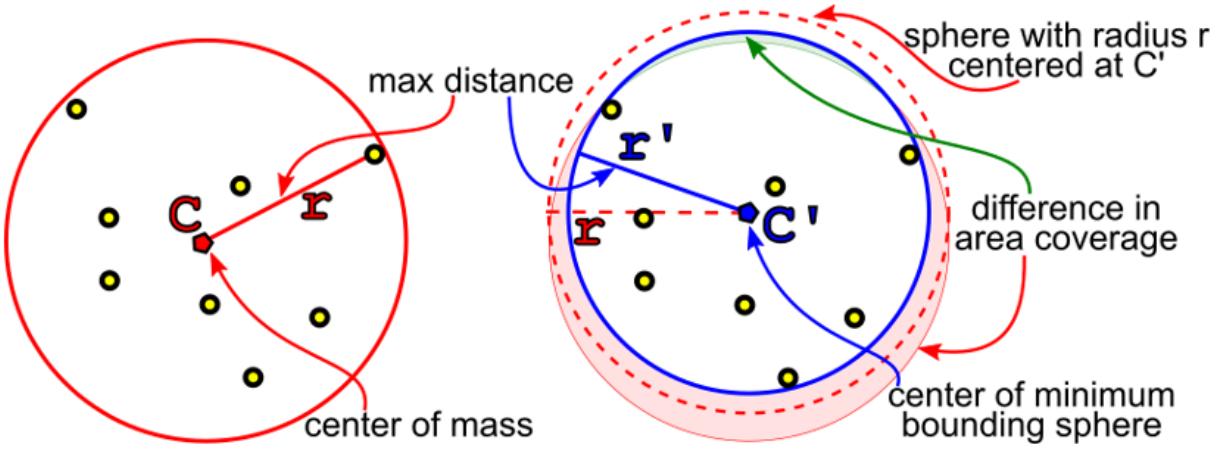


Figure 10.26 Difference between the spheres with minimum radius centered at the cluster centroid (left), and the minimum covering sphere (right) for an example set of points.